A4.6 Transaction buffering

Write accesses to the following memory types do not require a transaction response from the final destination, but do require that write transactions are made visible at the final destination *In a timely manner*:

- Device Bufferable
- Normal Non-cacheable Bufferable
- Write-through.

For write transactions, all three memory types require the same behavior. For read transactions, the required behavior is as follows:

- for Device Bufferable memory, read data must be obtained from the final destination
- for Normal Non-cacheable Bufferable memory, read data must be obtained either from the final destination or from a write transaction that is progressing to its final destination
- for Write-through memory, read data can be obtained from an intermediate cached copy.

In addition to ensuring that write transactions progress towards their final destination in a timely manner, intermediate buffers must behave as follows:

- An intermediate buffer that can respond to a transaction must ensure that, over time, any read transaction to Normal Non-cacheable Bufferable propagates towards its destination. This means that, when forwarding a read transaction, the attempted forwarding must not continue indefinitely, and any data used for forwarding must not persist indefinitely. The protocol does not define any mechanism for determining how long data used for forwarding a read transaction can persist. However, in such a mechanism, the act of reading the data must not reset the data timeout period.

  Note

  Without this requirement, continued polling of the same location can prevent the timeout of a read held in the buffer, preventing the read progressing towards its destination.

- An intermediate buffer that can hold and merge write transactions must ensure that transactions do not remain in its buffer indefinitely. For example, merging write transactions must not reset the mechanism that determines when a write is drained towards its final destination.

  Note

  Without this requirement, continued writes to the same location can prevent the timeout of a write held in the buffer, preventing the write progressing towards its destination.

For information about the required behavior of read accesses to these memory types, see:

- *Device Bufferable* on page A4-66
- *Normal Non-cacheable Bufferable* on page A4-66
- *Write-through No-allocate* on page A4-67.

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